ZARAGOZA. LOGISTICS CENTER

Product Wheel

MIT GLOBAL SCALE NETWORK

The problem: Many products, shared capacity

One asset must serve a varied product mix

Several products compete for the same line or machine.
Every switch takes time away from producing saleable output.

Especially relevant in process manufacturing

Chemicals and food often share processing or packaging assets.
Cleaning, material changes and pack-size changes can be costly.

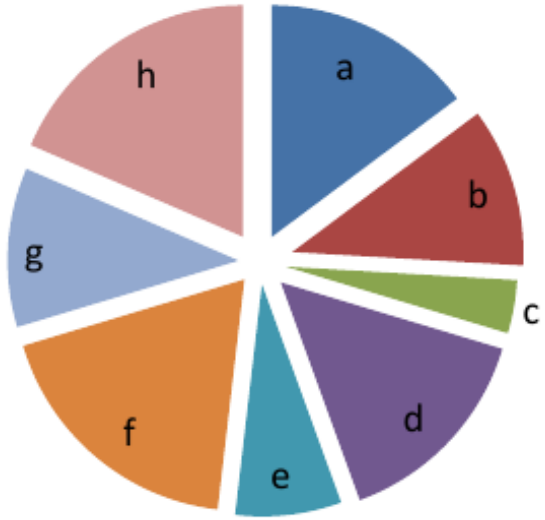
The scheduling challenge

Long campaigns save setups but build inventory for some items while others wait. Frequent switching consumes scarce capacity.

Aim: Meet demand while balancing inventory, setups and the productive use of shared capacity.

The production wheel

A repeating production sequence for several products that share the same machine or production line.



One turn = one wheel cycle

Each sector is a production run. The line switches between products and then repeats the sequence.

The cycle length determines how soon the line returns to a product.

Inventory covers demand until the next production run.

The wheel sets the production sequence and timing.

Designing the sequence and run frequencies

Group products to simplify changeovers

Choose a sequence around the important setup drivers, such as material type, cleaning needs or packaging size.

Bleach case: bottle-size change = 60 min, label change = 10 min. Grouping by bottle size limits the more time-consuming switches.

Give products suitable replenishment intervals

Make frequent sellers every cycle. Make other products every second or later cycle, or only when ordered.

Illustration: with a 2-day base cycle, one item runs every 2 days and another every 4 days. Quantities cover each item’s interval.

The EOQ connection: How often to produce

EOQ gives an economic lot-size benchmark for each product:

$$Q_i^* = \sqrt{(2S_iD_i / H_i)} \qquad T_i^* = Q_i^* / D_i$$

S_i = setup cost, D_i = demand rate, H_i = holding cost per unit per period
T_i^* = time between runs, in the same time unit used for demand

Shorter intervals mean smaller lots and less cycle inventory, but more setups. Longer intervals reverse this trade-off.

Choose a base cycle T and multiples k_i = 1, 2, 3, …
Then each production quantity covers demand: $Q_i = D_i \times k_iT$.

EOQ suggests intervals. The wheel must coordinate them within the capacity of the shared asset.

Capacity sets a limit on wheel frequency

Production and changeovers must fit the available time

More cycles mean smaller lots, but more time spent on setups. Better sequences or faster setups can release useful capacity.

Flour case: A weekly capacity check

Available time: 1,920 min. Production to meet demand: 1,650 min. Time left for changeovers: 270 min. Each wheel requires 180 min.

1 wheel/week: 1,650 + 180 = 1,830 min Fits
2 wheels/week: 1,650 + 360 = 2,010 min Exceeds capacity

The theoretical limit is 270 / 180 = 1.5 wheels per week. A workable schedule also needs room for demand and disruptions.